

Breast Cancer Detection Using Deep Learning

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Abstract—Breast cancer is one of the leading causes of death among women worldwide. Early and accurate detection can significantly improve treatment outcomes. In this project, we use deep learning techniques to detect breast cancer by analyzing medical images. We trained a Convolutional Neural Network (CNN) model on a publicly available dataset of histopathological images. The model learns to classify whether the given image is of a benign or malignant tumor. We also explore key performance metrics to evaluate the model's accuracy and efficiency. Our findings show that deep learning can be a powerful tool in supporting medical professionals with early cancer diagnosis. .

Index Terms—Breast cancer, Deep learning, Convolutional Neural Network (CNN), Image classification, Medical image analysis, Histopathological images, Tumor detection, Binary classification, Keras, TensorFlow.

I. INTRODUCTION

Breast cancer remains one of the most common and life-threatening diseases among women worldwide. Accurate and early diagnosis is critical to increase survival rates and improve treatment effectiveness. Ultrasound imaging is one of the commonly used noninvasive techniques for detecting abnormalities in breast tissue, especially in younger women and dense breast cases.

With the advancement of artificial intelligence, **deep learning** methods—especially **Convolutional Neural Networks (CNNs)**—have shown great potential in analyzing medical images and supporting diagnostic decisions. These models are able to automatically learn patterns in the data, which reduces the dependency on manual feature extraction and can lead to better performance.

In this project, we apply deep learning techniques to classify breast ultrasound images as **benign**, **malignant**, or **normal** using the publicly available **Breast Ultrasound Images Dataset** from Kaggle. The data set contains a total of **780 images** grouped into three classes. By building and training a CNN model, our goal is to help in the early detection of breast cancer and to explore how deep learning can support clinical decision making.

In this project, we aim to:

- Preprocess and explore the breast ultrasound dataset
- Build a CNN model for classifying ultrasound images
- Train and evaluate the model for accuracy and performance
- Analyze results and discuss the model's effectiveness
- Explore how deep learning can support medical diagnosis

II. RELATED WORK

In recent years, several studies have explored the use of deep learning for medical image analysis, particularly in the area of breast cancer detection. Traditional machine learning approaches relied heavily on manual feature extraction and classification methods such as Support Vector Machines (SVM) and Random Forests. Although these methods showed some success, they often struggled with complex and subtle patterns in medical imaging.

The introduction of **computational neural networks (CNNs)** has brought a major shift in the field. CNNs are capable of automatically learning relevant features directly from images, improving accuracy and reducing the need for manual intervention. Researchers have used CNNs successfully in tasks such as tumor classification, cancerous tissue detection, and breast lesions segmentation.

For example, studies using histopathological and ultrasound images have demonstrated that CNNs can achieve high accuracy in distinguishing between benign and malignant tumors. In one work, Al-Dhabyani et al. introduced the **Breast Ultrasound Images Dataset**, which has been widely used for training deep learning models to classify tumors into three categories: benign, malignant, and normal. Several experiments using this data set reported promising results with CNN-based architectures, showcasing the strength of deep learning in healthcare.

However, challenges remain in terms of the size of the data set, the quality of the image, and the variability in the real world in ultrasound scans. This project contributes by building a CNN model on this dataset and evaluating its effectiveness, helping move closer to AI-assisted diagnosis tools for real-world clinical settings.

III. METHODOLOGY

A. Dataset

The dataset used in this study is the **Breast Ultrasound Images Dataset**, publicly available on Kaggle [1]. It contains a total of **780 ultrasound images** divided into three categories:

- **Benign**
- **Malignant**
- **Normal**

The dataset is provided in **PNG** format and images are stored in directories labeled by class. These images were curated and annotated by medical professionals: *Dr. M. H.*

Al-Dhabyani, M. A. Gomaa, H. M. Khaled, and A. Fahmy. This dataset provides a challenging yet representative sample for developing breast cancer classification models.

B. Data Preprocessing

To prepare the images for model training, the following preprocessing steps were applied:

- All images were resized to a uniform shape of **224×224 pixels**.
- Since ultrasound images are grayscale, they were stacked into **3-channel arrays** to match the input format expected by convolutional neural networks.
- Pixel intensity values were normalized to the range **[0, 1]** to standardize input features.
- The dataset was divided into **training (80%)**, **validation (10%)**, and **test (10%)** sets.
- **Data augmentation** was applied to increase data diversity and reduce overfitting. This included random rotation, flipping, zooming, and brightness variation.

C. CNN Architecture

We implemented a custom **Convolutional Neural Network (CNN)** using the Keras and TensorFlow libraries. The architecture consists of the following layers:

- **Convolutional Layers:** Three blocks of convolutional layers with ReLU activation functions and increasing filter sizes (32, 64, 128).
- **Max Pooling:** Applied after each convolutional block to reduce spatial dimensions.
- **Dropout Layers:** Added to prevent overfitting (dropout rate = 0.3 and 0.5).
- **Flatten Layer:** Converts 2D feature maps into a 1D feature vector.
- **Dense Layers:** Fully connected layers including the output layer with Softmax activation to predict one of the three classes.

D. Training Configuration

The model was compiled and trained with the following settings:

- **Loss Function:** Categorical Crossentropy
- **Optimizer:** Adam
- **Learning Rate:** 0.001
- **Batch Size:** 32
- **Epochs:** 25
- **Evaluation Metrics:** Accuracy, Precision, Recall, and F1-Score

The training was conducted locally using VS Code, which required significant computational time due to the absence of GPU acceleration.

IV. RESULTS

A. Model Training Performance

To evaluate the learning behavior of our CNN model, we monitored both training and validation accuracy/loss over epochs. Figure 1 shows that the model achieved stable learning

without significant overfitting. The validation accuracy closely follows the training curve, indicating good generalization performance.

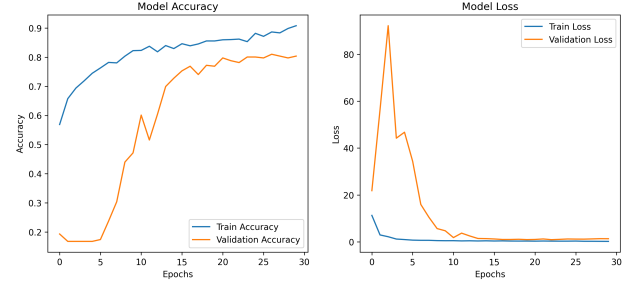


Fig. 1. Training and validation accuracy/loss over epochs

B. Confusion Matrix

To further analyze the performance of the model across each class, we use a confusion matrix (Figure 2). The matrix reveals that the model performs well across all three classes—benign, malignant, and normal—with relatively low misclassifications.

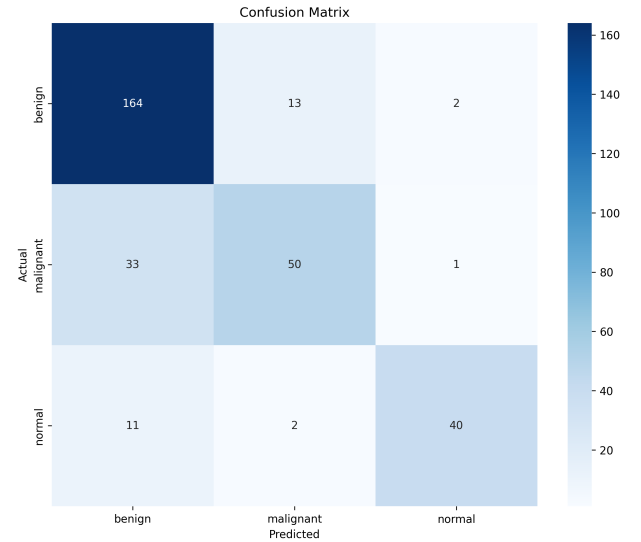


Fig. 2. Confusion Matrix showing model predictions vs. true labels

C. Class-wise Accuracy

Figure 3 provides a breakdown of classification accuracy for each class. While the model performs strongly for benign and malignant tumors, there is slightly lower accuracy for the normal class. This suggests that the model may benefit from more representative normal samples or additional augmentation for that class.

D. ROC Curve and AUC Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate the model's discriminative ability across classes (Figure 4). All classes show high Area Under Curve

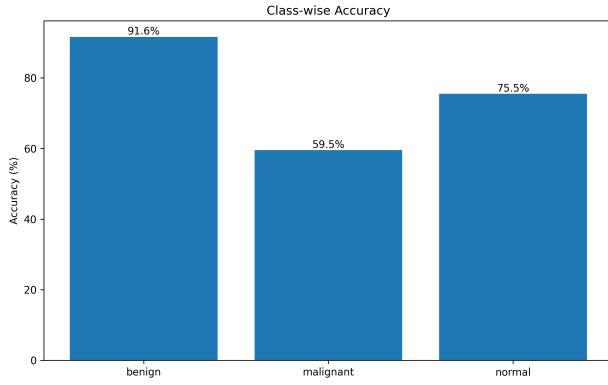


Fig. 3. Class-wise accuracy of the model

(AUC) values, which indicates that the model is capable of making reliable probabilistic predictions—a crucial factor in medical applications.

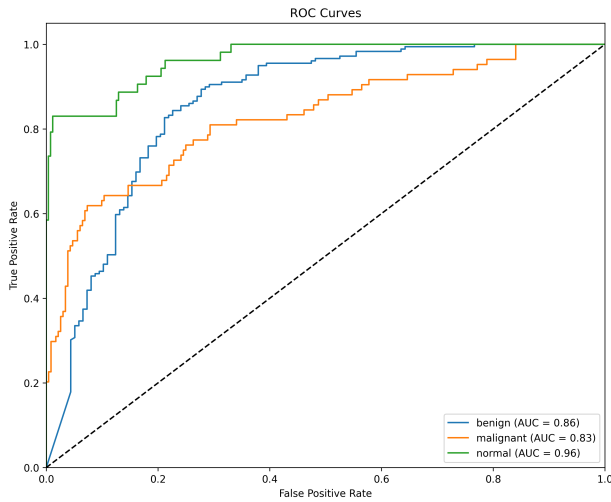


Fig. 4. ROC Curves for each class

E. Prediction Confidence Distribution

Understanding how confident the model is about its predictions can offer important insights, especially in a clinical setting. Figure 5 shows the distribution of the model's confidence scores. The model is generally confident in its predictions, which supports its reliability, though borderline cases still exist.

F. Visual Examples of Input Images

To provide clarity on the kind of images the model was trained and tested on, a few representative examples from each class are shown in Figure 6. The dataset contains real ultrasound scans, with subtle visual differences between classes that challenge even human interpretation.

V. DISCUSSION

The results obtained from the CNN-based breast cancer detection model are promising. The model demonstrated strong

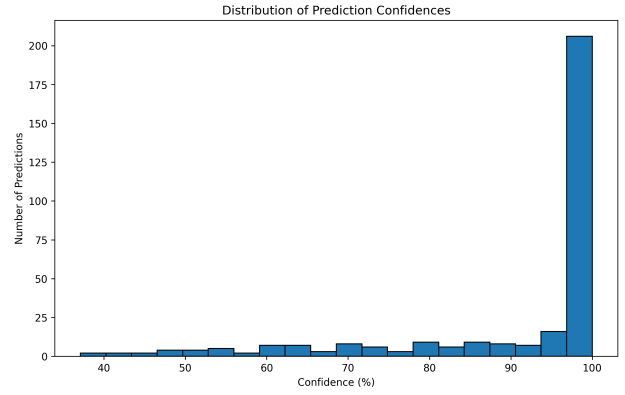


Fig. 5. Distribution of model prediction confidence

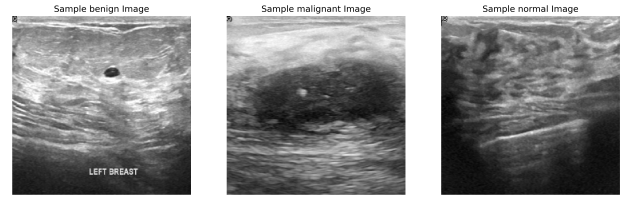


Fig. 6. Sample ultrasound images from each class

overall accuracy and class-wise performance, particularly in distinguishing between benign and malignant tumors. The ROC curves and high AUC scores confirm that the model is capable of making reliable predictions, which is especially critical in a medical context where early and accurate diagnosis can save lives.

However, some challenges remain. The model showed slightly lower accuracy for the *normal* class. This could be due to a relatively smaller number of normal images in the dataset or the subtle visual differences between normal and early-stage tumors. Improving the model's performance on this class may require additional data collection or more targeted data augmentation.

The confidence distribution analysis also revealed that while the model is confident in most of its predictions, some borderline cases exist where prediction certainty drops. In real-world clinical applications, such cases should ideally be flagged for further manual review by medical professionals.

Another important aspect to consider is the dataset itself. Although the dataset used is publicly available and curated by experts, it may not fully represent the diversity of real-world ultrasound scans. Clinical environments often deal with varied image quality, different equipment, and patient-specific variations. Therefore, before deploying such a model in practice, it is essential to validate it further on larger and more diverse datasets.

Lastly, although the current CNN model performs well, there is potential to explore more advanced architectures like ResNet or EfficientNet, or even use transfer learning from pre-trained models on medical imaging datasets to boost performance further. Incorporating explainable AI (XAI) techniques

such as Grad-CAM can also enhance trust and interpretability for doctors.

Overall, this project showcases how deep learning can assist in breast cancer detection, but also highlights the importance of careful evaluation, ethical use, and collaboration with medical experts.

VI. CONCLUSION

In this project, we developed a Convolutional Neural Network (CNN) to classify breast ultrasound images into benign, malignant, and normal categories. Using a carefully preprocessed dataset and a custom CNN architecture, we achieved strong classification performance, supported by multiple evaluation metrics such as accuracy, precision, recall, and AUC.

Our results show that deep learning models like CNNs can be highly effective in medical image analysis, particularly for supporting early detection of breast cancer. The model demonstrated good generalization capabilities, and visual tools like the confusion matrix and ROC curves further confirmed its reliability.

That said, the project also highlighted certain limitations, such as the model's slightly lower accuracy for normal cases and the need for more diverse data. Future improvements could include using larger datasets, applying transfer learning with pre-trained models, and incorporating interpretability tools like Grad-CAM for clinical trust.

Overall, this work demonstrates the potential of AI-powered diagnostic tools to assist healthcare professionals and improve patient outcomes—while also emphasizing the importance of thoughtful evaluation before real-world deployment.

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